Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · INDEPENDENT T-TEST

do two groups' means differ?

Table 1. Group statistics

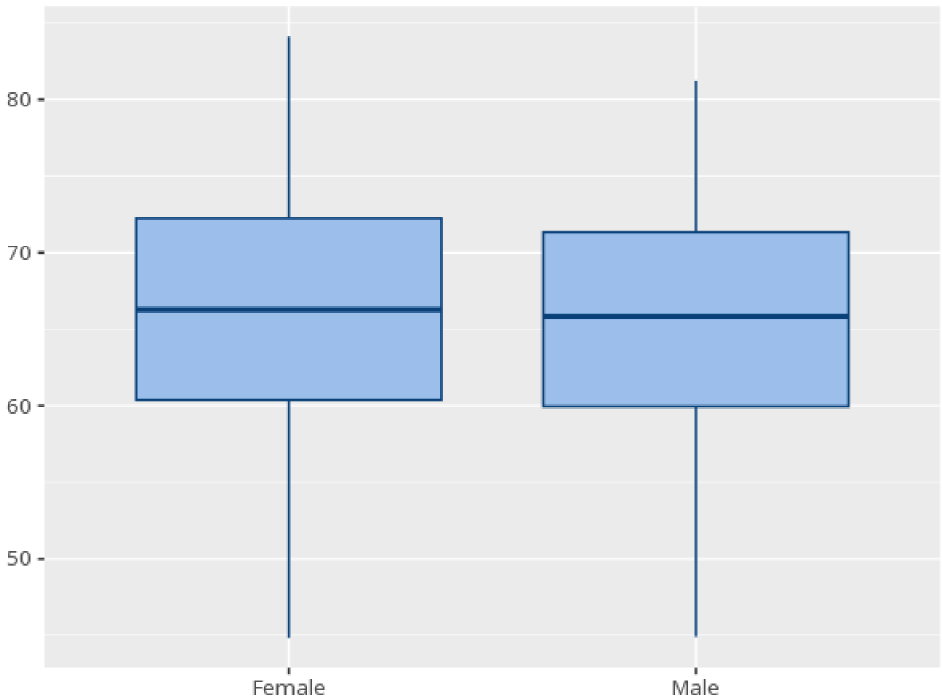
Group	N	Mean	Std. Dev.	SE
Female	120	65.69	8.31	0.76
Male	120	65.44	8.03	0.73

Table 2. Independent-samples t-test

Contrast	t	df	p	M_{diff}	95% CI	Cohen's d [95% CI]
Female – Male	0.24	237.73	.810	0.25	[−1.82, 2.33]	0.03 [−0.22, 0.28]

assumption checks: Levene's test for equal variances & within-group normality (Shapiro-Wilk per group); a Welch row replaces the pooled row when equal-variance is off. (Levene F=0.27, p=.605 · Shapiro Female W=0.99, p=.474; Shapiro Male W=0.99, p=.466 · Welch test) — equal variances look reasonable — normality looks reasonable in both groups

Figure. Distribution of the outcome by group



Understanding the numbers

t. How many standard errors the two group means are apart. Here, t(237.73) = 0.24.

p. The probability of a difference this large (or larger) if the groups truly had equal means. Here, p = .810 - at or above the conventional .05 threshold, no significant difference detected.

Cohen's d. The size of the gap in standard-deviation units (~0.2 small, 0.5 medium, 0.8 large). Here, d = 0.03 [−0.22, 0.28].

N. The number of complete cases (rows) in each group. Here, n = 120 for Female and n = 120 for Male.

Mean. The average outcome value in each group. Here, M = 65.69 for Female and M = 65.44 for Male.

SD. How spread out the outcome values are around each group's mean. Here, SD = 8.31 for Female and SD = 8.03 for Male.

SE. The precision of each group's mean as an estimate of its population mean. Here, SE = 0.76 for Female and SE = 0.73 for Male.

df. The degrees of freedom for the t-test (pooled or Welch-adjusted). Here, df = 237.73.

M diff. The difference between the two group means. Here, the mean difference is 0.25.

95% CI. The range of mean differences plausible at the chosen confidence level, given this sample. Here, the 95% CI for the mean difference is [−1.82, 2.33].

How to read this test

Asks whether two groups' averages differ by more than chance. Look at p: below alpha (e.g. 0.05) means significant. The mean difference and 95% CI show the size and precision of the gap; Cohen's d gives effect size (~0.2 small, 0.5 medium, 0.8 large). A large p means no significant difference was detected — not that the group means are equal.

APA template: An independent-samples t-test compared Female (M=65.7, SD=8.3) and Male (M=65.4, SD=8.0), t(237.73)=0.24, p = .810, d=0.03 [−0.22, 0.28].

Statistical basis: Independent-samples t-test (pooled variance) (Student (1908). "The probable error of a mean." Biometrika, 6(1), 1-25.); Welch's correction (unequal variances, the app's default) (Welch, B. L. (1947). "The generalization of Student's problem when several different population variances are involved." Biometrika, 34(1-2), 28-35.); Cohen's d effect-size benchmarks (.2/.5/.8) (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.); Levene's test for equal variances (Brown-Forsythe median-centered variant) (Levene, H. (1960). "Robust tests for equality of variances." In I. Olkin (Ed.), Contributions to Probability and Statistics (pp. 278-292). Stanford University Press.); Brown-Forsythe median-centered variant of the homogeneity-of-variance test (Brown, M. B., Forsythe, A. B. (1974). "Robust tests for the equality of variances." Journal of the American Statistical Association, 69(346), 364-367.). Full references in the exported CITATIONS.txt.